

HONGSHAN CHEN

Mobile:(+65) 9195 4027

Email: e1471281@u.nus.edu

[GitHub](#)

[LinkedIn](#)

[Portfolio](#)

EDUCATION

National University of Singapore

Aug 2024 - Present

Bachelor of Computing in Computer Science

- Scholarship: NUS Science and Technology Scholarship
- Academic courses: Programming Methodology II (A module focusing on object-orientated programming, using Java), Data Structure and Algorithms, Computer Organisation, Computer Graphics

EXPERIENCE

NetEase Cloud Music CS Agent (Individual Project), *Individual*

Dec 2025 - Jan 2026

- Built a runnable customer-support Agent backend with a RAG pipeline (retrieve → evidence injection → generate), optimised for reproducible demos.
- Implemented REST APIs with Spring Boot + Swagger; persisted the knowledge base in MySQL using JPA Entities/Repositories and added Top-K evidence retrieval for prompt grounding.
- Added a strict Refusal Gate (hits = 0 → refuse, no LLM call) to prevent ungrounded outputs and reduce hallucination risk.
- Introduced Redis caching with TTL and graceful degradation; configured Spring Profiles + docker-compose (MySQL/Redis) for dev/prod separation and one-command startup.
- Practiced AI-assisted (vibe) coding with Cursor/LLM for scaffolding, refactoring, and docs iteration, with manual review and local verification for core logic.
- [Project link](#)

TikTok TechJam 2025 (Singapore), *Participant*

Aug 2025 - Sep 2025

- Built a prototype system that filters and categorizes Google Reviews across multiple databases using LLMs + multimodal models, removing invalid/low-quality content to improve data usability.
- Led the model architecture design and implemented a layered multimodal hybrid pipeline combining FastBERT, ChatGPT-4o, and Qwen3-8B. Integrated CLIP for text-image alignment to improve multimodal consistency and boost review-filtering accuracy.
- Applied prompt engineering to strengthen instruction following and reduce misclassification on ambiguous cases.
- Improved robustness against cross-database schema/label inconsistencies and semantic ambiguity via hierarchical decision design.
- [Project link](#)

OpenGL Graphics Coursework (Individual Project), *Individual*

Aug 2025 - Dec 2025

- Implemented core rendering features including MVP transforms, lighting and shading, texture mapping, shadow rendering and ray tracing, with per-effect toggles for demos/debugging.
- Delivered a demo-ready project page with recorded showcases and clear documentation of controls and rendering pipeline.
- [Project link](#)

SKILLS SUMMARY

- **Programming Languages:** Java (advanced); C, C++, Python, HTML/CSS, JavaScript, Markdown (basic)
- **Frameworks / Libraries:** Spring Boot, OpenGL, Hugging Face Transformers, PyTorch (basic)
- **AI / Machine Learning:** Familiar with prompt engineering using IDE such as Cursor; Familiar with Large Language Models (LLMs) such as ChatGPT, Qwen, and BERT; Experience with multimodal models such as CLIP
- **Tools / Platforms:** Git, GitHub; JetBrains IDEs; Aseprite; Microsoft Office (Word / PowerPoint / Excel)

EXTRA CURRICULAR ACTIVITIES

NUS Piano Ensemble, *Member*

Aug 2024 - Present

- Conducted 4 duets with partner in annual concerts and participated actively in club's media and publication matters

NUS Climbing Club, *Member*

Aug 2025 - Present

- Hold Level 2 Certificate of Sport Climbing (SNCS 2) in Singapore

Chengdu Shude High School, *Principal Pianist*

Jan 2022 - Jan 2024

- Conducted and led 6 concerts as the pianist of the school's orchestra and chorus, also presenting 5 more solo pieces